MSG No. 133.465 $P4_2/nb'c'$ [Type III, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `22'2'`.

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,4,9,10]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,3,11,12]`
3	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[5,8,13,14]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,15,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `22'2'`.

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,4,9,10]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,11,12]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[5,8,13,14]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[6,7,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,4,11,12]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,3,9,10]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,8,15,16]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,7,13,14]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,4,6,7]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,5,8]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[11,12,13,14]`

Table 5: Wyckoff site: `8e`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,6]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8]`
5	$\left[0, \frac{1}{2}, 0\right]$	`[9,13]`
6	$\left[\frac{1}{2}, 0, 0\right]$	`[10,14]`
7	$\left[0, 0, \frac{1}{2}\right]$	`[11,15]`

8	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[12,16]

Table 6: Wyckoff site: 8f, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[1,4]
2	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[2,3]
3	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	[5,8]
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[6,7]
5	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	[9,10]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[11,12]
7	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	[13,14]
8	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: 2..

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[1,4]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[2,3]
3	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[5,8]
4	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[6,7]
5	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[9,10]
6	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[11,12]
7	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[13,14]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[15,16]

Table 8: Wyckoff site: 8h, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ 0\right]$	[1,9]
2	$\left[\frac{1}{4},\ x,\ \frac{1}{2}\right]$	[2,11]
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[3,12]
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$	[4,10]
5	$\left[-x,\ \frac{3}{4},\ 0\right]$	[5,13]
6	$\left[\frac{3}{4},\ -x,\ \frac{1}{2}\right]$	[6,15]
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$	[8,14]

Table 9: Wyckoff site: 8i, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[1,9]
2	$\left[\frac{1}{4},\ x,\ 0\right]$	[2,11]
3	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ 0\right]$	[3,12]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	[4,10]
5	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	[5,13]
6	$\left[\frac{3}{4},\ -x,\ 0\right]$	[6,15]
7	$\left[\frac{3}{4},\ x+\frac{1}{2},\ 0\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	[8,14]

Table 10: Wyckoff site: 8j, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	[1,11]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[2,10]
3	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[3,9]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[4,12]
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[5,15]
6	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[6,14]
7	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[7,13]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[8,16]

Table 11: Wyckoff site: 16k, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	[2]
3	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[6]
7	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[7]
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[8]
9	$\left[x,\ \frac{1}{2}-y,\ -z\right]$	[9]
10	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[10]
11	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[12]
13	$\left[-x,\ y+\frac{1}{2},\ z\right]$	[13]
14	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[14]
15	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[16]